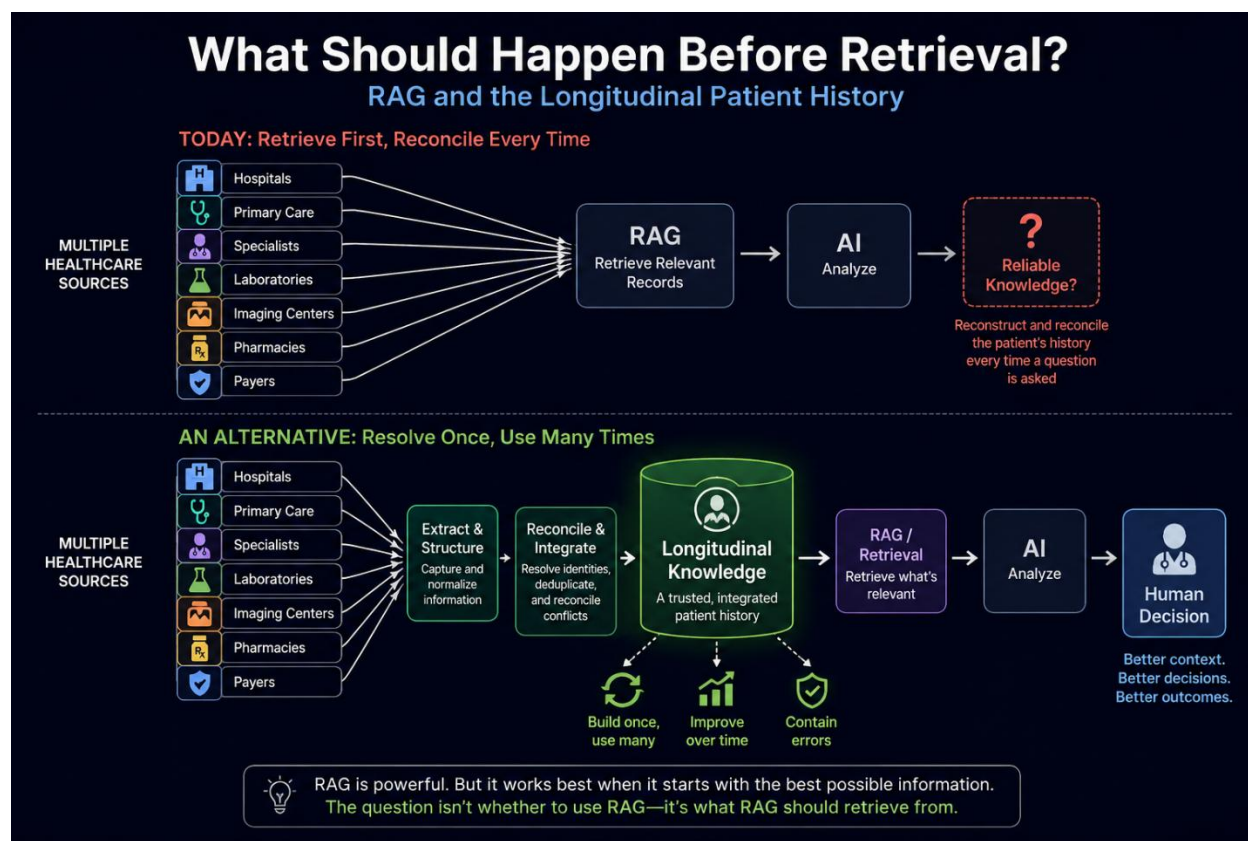


Does RAG Solve Healthcare AI's Information Problem?



Healthcare Data → AI ≠ Reliable Healthcare Knowledge

That was one conclusion I reached while developing the Future Healthcare Information Architecture.

But what about:

Healthcare Data → RAG → AI = Reliable Healthcare Knowledge?

RAG retrieves information relevant to a question. But retrieval doesn't resolve information that is inconsistent, duplicated, incomplete, or scattered across healthcare organizations.

I learned this lesson in 1996 while analyzing the enterprise data warehouse of a \$10 billion company with roughly 60,000 employees. Data had been loaded without sufficient cleaning and integration, leaving those problems to report developers.

Contractors used trial and error to determine how tables should be joined and interpreted. **Because the source data had not been resolved, different analysts could get different answers to the same question, while report development consumed roughly one-third of the warehouse capacity.**

The problem wasn't the reporting tools—the information needed to be resolved before it was used.

Healthcare faces the same issue with RAG.

A patient's history can span decades and many healthcare organizations. **Interoperability is making that information increasingly accessible. But accessible information can still conflict, duplicate earlier information, describe unsuccessful treatments or suspected conditions, and contain later evidence that changes the interpretation of earlier events.**

RAG can retrieve those records. **But if the information hasn't been resolved, AI could reach different answers depending on how it interprets conflicts. Should AI reconstruct and reconcile the history every time someone asks a question?**

A longitudinal patient history takes a different approach:

Healthcare Data → Extract & Structure → Reconcile & Integrate → Longitudinal Knowledge

Then, when needed:

Longitudinal Knowledge → Retrieve → AI → Human Decision

This applies an old data-architecture principle: **Build once, use many.**

Efficiency. Don't repeatedly reconstruct what has already been learned, reducing processing, cost, and response time.

Improvement. When information conflicts, AI can identify it and a practitioner—or, when appropriate, the patient—can resolve it. The resolution persists while the original evidence and provenance remain available.

Error containment. Separating extraction and reconciliation from retrieval and analysis provides validation points and limits how far an error can propagate. Correct the longitudinal history once rather than repeatedly influencing future analyses.

RAG can make AI better at finding information. A longitudinal architecture can improve the information itself.

RAG remains part of the solution. But before asking **“How do we retrieve more of the patient's records?”**, perhaps healthcare AI should first ask:

“What should happen to those records before we retrieve them?”

FHIA-08 is part of the Future Healthcare Information Architecture series:

doungebel.github.io/HealthcareAI